

3D reconstruction of spatial transcriptome

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1 Motivation

Spatial transcriptomics (ST) has greatly advanced our understanding of tissue organization, but building comprehensive three-dimensional (3D) cell atlases remains challenging due to technical constraints and high experimental costs.

cf) 3D omics platform, Pyxa <https://stellaromics.com/product/>

Most current ST technologies operate on two-dimensional (2D) tissue sections. A common strategy to obtain 3D information is to acquire multiple serial sections and stack them along the z-axis to form a volumetric representation. In practice, however, obtaining sufficiently dense and well-aligned serial sections is often infeasible. The cost of ST experiments scales linearly with the number of sections, while technical issues such as tissue deformation further limit achievable z-resolution. As a result, existing z-stack-based 3D reconstructions remain sparse along the depth dimension and fail to capture a complete 3D cell atlas. In this work, we address this limitation by interpolating the missing ST sections between sparsely sampled sections, rather than relying on densely sampled serial sections, thus reconstruct 3D cell atlas.

2 Notation

Given a set of n serial ST sections from a single patient, we denote the observed data as

$$\mathbf{X} = \{X^{(i)}\}_{i=1}^n, \quad X^{(i)} \in \mathbb{R}^{N_i \times G}, \quad (1)$$

where each $X^{(i)}$ represents the i -th ST section consisting of N_i spots/cells and G genes. Each section is associated with a depth coordinate $z_i \in \mathbb{R}$,

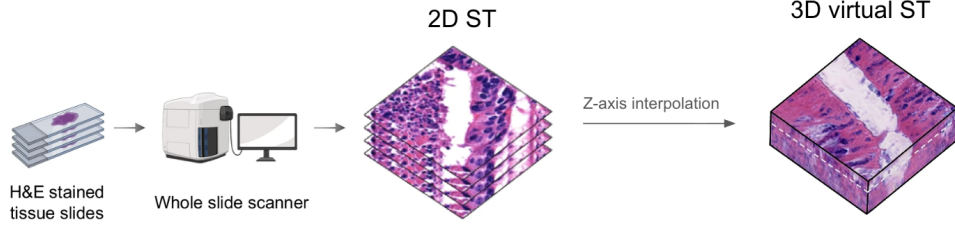


Figure 1: Motivation

reflecting its position along the sectioning axis, with sections ordered such that $z_i < z_{i+1}$.

Each spot/cell within section i is associated with a two-dimensional spatial coordinate

$$\mathbf{P}^{(i)} \in \mathbb{R}^{N_i \times 2}, \quad (2)$$

where $\mathbf{P}_j^{(i)} = (x_j^{(i)}, y_j^{(i)})$ denotes the spatial location of the j -th spot/cell in section i .

Together, each section can be represented as a set of spatially resolved gene expression measurements,

$$X^{(i)} = \{(X_j^{(i)}, \mathbf{P}_j^{(i)}, z_i)\}_{j=1}^{N_i}, \quad (3)$$

where $X_j^{(i)} \in \mathbb{R}^G$ denotes the gene expression vector of the j -th spot/cell.

3 Problem definition

We formulate the 3D ST reconstruction problem as a **3D image rendering** problem given a set of 2D observations (i.e. 2D static scene). In classical 3D reconstruction pipelines, such as Structure-from-Motion (SfM) [6], a scene is first represented by a **sparse** three-dimensional geometric structure, typically in the form of a point cloud, where each point corresponds to a distinct visual feature triangulated from multiple views. Although sparse, this representation captures the global spatial layout of the scene and serves as a foundation for subsequent dense reconstruction.

Analogously, 3D ST can be naturally interpreted as a sparse 3D geometric representation of tissue. Given an ordered set of serial ST sections $\{X^{(i)}\}_{i=1}^n$, each spot/cell provides a spatial location (x, y) within a section

and an associated depth z_i , together with a high-dimensional gene expression vector. This allows the observed data to be represented as a gene-annotated 3D point cloud,

$$\mathcal{P}_{\text{ST}} = \{(\mathbf{p}_j, \mathbf{g}_j) \mid \mathbf{p}_j = (x_j, y_j, z_j) \in \mathbb{R}^3, \mathbf{g}_j \in \mathbb{R}^G\}, \quad (4)$$

where each point corresponds to an individual ST spot/cell. Due to the limited number of sections and spots/cells, this point cloud is inherently sparse along the depth dimension, motivating the reconstruction of a continuous 3D tissue representation from sparse observations.

Rather than explicitly interpolating missing sections along the depth axis, we model spatial gene expression as a continuous three-dimensional field

$$f : \mathbb{R}^3 \rightarrow \mathbb{R}^G \quad (5)$$

where the function returns a G -dimensional gene expression vector at spatial location (x, y, z) . Observed ST sections correspond to samples of this field at discrete depths z_i , such that

$$X_j^{(i)} \approx f(x_j^{(i)}, y_j^{(i)}, z_i). \quad (6)$$

A two-dimensional section at an arbitrary depth z is obtained by rendering the continuous field through a rendering operator $\mathcal{R}$,

$$\hat{X}(x, y; z) = \mathcal{R}(f, z). \quad (7)$$

In this formulation, interpolation between sparsely sampled sections emerges implicitly by querying the learned continuous representation at unseen depths, so that rendering itself serves as a principled mechanism for interpolating gene expression across space and reconstructing a continuous 3D tissue architecture from sparse observations.

Q.H&E image is not included at the moment. How to incorporate it? Is it necessary? A. H&E is easy-to-require modality. There's clear usecase for translation from H&E to ST. One option can be.. 'splat' on H&E yields ST!

Q.Since ST is measured in single camera direction (i.e. z-stack).

4 Preliminary

"Existing researches have challenged the prevailing assumption that images are best represented as uniform pixel grids given the continuous nature of real visual world" (Should be paraphrased) [8].

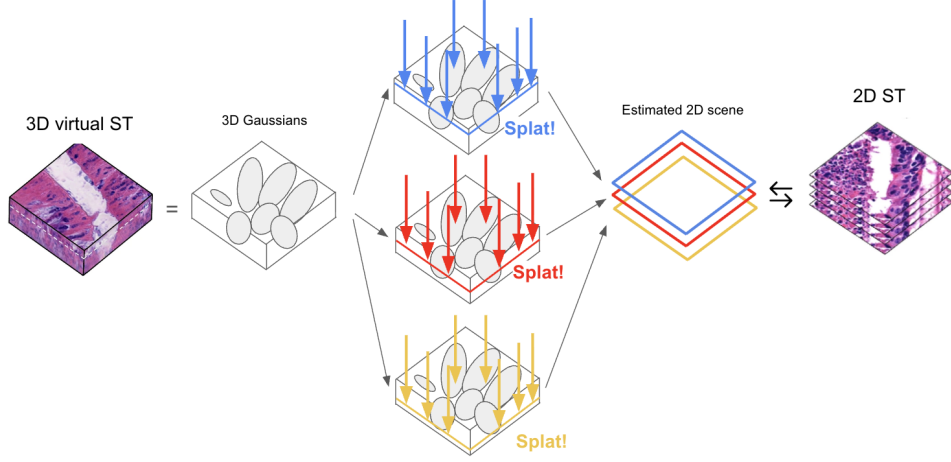


Figure 2: Schematic overview

4.1 Point-cloud representation of scene

Structure-from-Motion (SfM)[6]

4.2 Gaussian splatting

Gaussian splatting represents a virtual 3D scene as a collection of Gaussians, which are projected (i.e. **splatted**) into 2D to generate a static scene. Under this assumption, the method reconstructs the 3D scene by iteratively minimizing the discrepancy between the observed 2D images and the estimated 2D renderings [2].

The following describes the Gaussian splatting method in detail. Starting from a sparse set of SfM points, a collection of three-dimensional Gaussians (3D Gaussians) is constructed. Each Gaussian is differentiable and can be easily projected into image space as a 2D splat, enabling the rendering of 2D images. A 3D Gaussian is defined by a mean $\mu \in \mathbb{R}^3$ and a full three-dimensional covariance matrix $\Sigma \in \mathbb{R}^{3 \times 3}$, and is expressed as

$$G(\mathbf{x}) = \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^\top \Sigma^{-1}(\mathbf{x} - \mu)\right). \quad (8)$$

Zwicker et al. [9] describe how such 3D Gaussians can be projected for rendering. Given a viewing transformation W , the covariance matrix in camera

coordinates is given by

$$\Sigma' = JW\Sigma W^\top J^\top, \quad (9)$$

where J denotes the Jacobian of the projective transformation. Since the covariance matrix Σ of a 3D Gaussian is equivalent to describing the shape of an ellipsoid, it can be parameterized using a scaling matrix S and a rotation matrix R as

$$\Sigma = RSS^\top R^\top. \quad (10)$$

The scaling and rotation parameters are optimized independently. The optimization is performed through successive iterations of rendering and comparing the resulting images to the training views in the captured dataset.

4.3 NeRF vs Gaussian splatting

Assume we have 2D static images captured from multiple viewpoints. NeRF [3] assumes the existence of a “virtual camera” and parameterizes the camera position and radiance so that, after training, it can render a 2D static image from an arbitrary camera viewpoint.

5 Experiments

5.1 Baselines

- ASIGN [7]
- HoloTea [4]: Not continuously/smoothly interpolate z-stack. (Inference time) H&E section is required to do ST prediction.
- UniST [5]
- VorTex [1]

5.2 Dataset

Same dataset from ASIGN

5.3 Evaluation metric.

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